

Srinivasarao Cherukupalli

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SUMMARY

Junior AI & Python Developer with expertise in designing and deploying end-to-end AI Voice Agents, RAG pipelines, and automated workflows using FastAPI, LLMs, and modern ML toolchains.

EXPERIENCE

VMax e-Solutions India Pvt. Ltd

Nov 2025 – Present

Associate AI Engineer

(On-Site) Hyderabad, Telangana

- Designing end-to-end AI Voice Agents for customer service workflows using LLMs and RAG pipelines.
- Building intelligent call-handling agents capable of understanding customer queries, generating context-aware responses, and completing tasks autonomously.
- Implementing Retrieval-Augmented Generation (RAG) to improve accuracy, reduce hallucinations, and provide real-time knowledge-grounded responses.
- Developing and fine-tuning machine learning models to optimize agent behaviour, improve call quality, and enhance customer experience.
- Integrating Pipecat AI pipelines with backend services, APIs, and telephony platforms (Twilio, SIP)

Staffbee Solutions INC

July 2025 – Sep 2025

Software Developer Intern

(On-Site) Hyderabad, Telangana

- Developed Python-based automation frameworks using Playwright to streamline repetitive web workflows, reducing manual effort by 80%.
- Designed and implemented robust test scripts for dynamic web applications, ensuring cross-browser compatibility and faster execution.
- Applied error-handling, logging, and parallel execution strategies to enhance stability and performance.

Codegnan IT Solutions

Dec 2024 – July 2025

Jr Python Developer

(On-Site) Hyderabad, Telangana

- Redesigned the company website using HTML, CSS, and JavaScript to improve user experience.
- Delivered hands-on guidance on building full-stack applications using Flask with real-time projects.
- Coordinated with project mentors and teammates to prioritize features and implement backend solutions in agile sprints.

PROJECTS

Building AI Voice Agent, Chatbot's Using Local Models

Jan 2026 – Present

Python, FastAPI, ASR, LLM, TTS, RAG, Pipecate

- Engineered end-to-end conversational voice agents utilizing local LLMs and real-time audio components.
- Integrated speech-to-text (ASR) and text-to-speech (TTS) frameworks to achieve seamless, low-latency spoken interactions.
- Constructed RAG pipelines backed by vector databases to ground agent responses in enterprise data.
- Implemented FastAPI backend routing to handle concurrent stream sessions and webhook integrations with telephony platforms.
- Containerized agent services using Docker to facilitate consistent deployment and scaling across cloud environments.

Image Extraction (OCR) Using Local models (Paddle, Doctr)

Nov 2025 – Jan 2026

Python, FastAPI, JavaScript, Regx, Paddle OCR, Doctr OCR

- Deployed local OCR models (PaddleOCR and Doctr) to extract high-accuracy structured text from complex documents and scanned images.
- Built high-performance FastAPI endpoints to process document uploads and return structured JSON outputs asynchronously.

- Applied advanced Regular Expressions (RegEx) and post-processing algorithms to clean, parse, and validate extracted text data.
- Integrated logging and error-handling utilities to ensure high uptime and capture edge cases during data extraction.

Web Scraping Using Playwright

July 2025 – Sep 2025

Python, JavaScript, FastAPI, Playwright, MySQL

- Designed and implemented robust test scripts for dynamic web applications, ensuring cross-browser compatibility and faster execution.
- Integrated automation scripts with FastAPI and SQL databases for data-driven execution and reporting.
- Optimized automation workflows by implementing selector best practices and dynamic waits, improving script reliability.
- Engineered error recovery and retry protocols to handle unexpected browser state changes during heavy scraping runs.
- Structured data storage pipelines to securely log extracted data directly into relational databases.

TECHNICAL SKILLS

Programming & AI/ML : Python, Pandas, NumPy, scikit-learn, PyTorch, Flask, REST APIs, SQL, HTML, CSS, JS

GenAI & Databases : Prompt Engineering, RAG concepts, MySQL, PgSQL

Testing & MLOps : Playwright, Docker, Git/GitHub, CI/CD (basic), Postman, Agile

Concepts : Machine Learning Basics (Regression, Classification, Clustering), SDLC, OOPs, DSA, Cloud (GCP basics)

EDUCATION

Guntur Engineering College

Sep 2020 – May 2024

Bachelor of Technology (CSE) CGPA – 8.29

Guntur, AP

CERTIFICATIONS

- Python Full Stack Developer – Pantech e-Learning
- Python, SQL (Basic & Advanced) – Hacker Rank